

# Solutions: Assignment 1

Course: Probability and Statistics

## Q1, 10 points

Given  $A = \{2, 4, 6\}$  and  $B = \{1, 2, 3, 4\}$  with  $S = \{1, 2, 3, 4, 5, 6\}$ .

$$\begin{aligned} A \cup B &= \{x \in S : x \in A \text{ or } x \in B\} \\ &= \{1, 2, 3, 4, 6\}. \end{aligned}$$

$$\begin{aligned} A \cap B &= \{x \in S : x \in A \text{ and } x \in B\} \\ &= \{2, 4\}. \end{aligned}$$

$$A^c = S - A = \{1, 3, 5\}.$$

*Grading idea (10 pts):* union (3.5), intersection (3.5), complement (3).

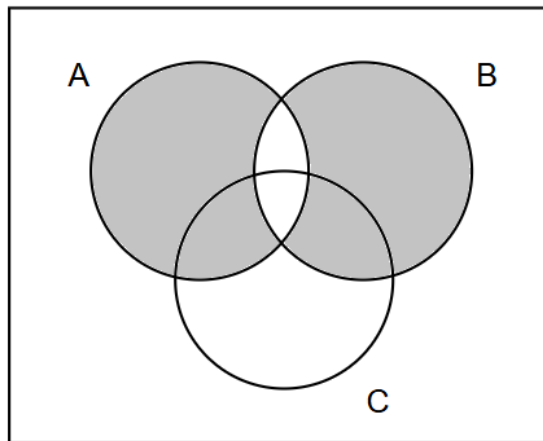
## Q2, 10 points

Let  $A = \{\text{heart}\}$ ,  $B = \{\text{face (J,Q,K)}\}$ ,  $C = \{\text{ace}\}$ .

1. “heart or face but not both” is the symmetric difference:

$$A \cup B - A \cap B \text{ or, } (A - B) \cup (B - A).$$

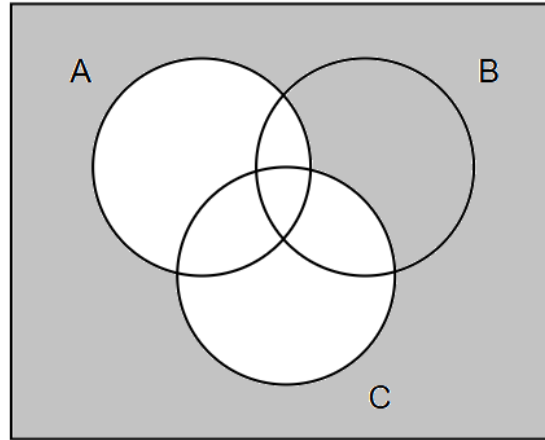
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2. “not a heart and not an ace” is the intersection of complements:

$$A^c \cap C^c \text{ or, } (A \cup C)^c.$$

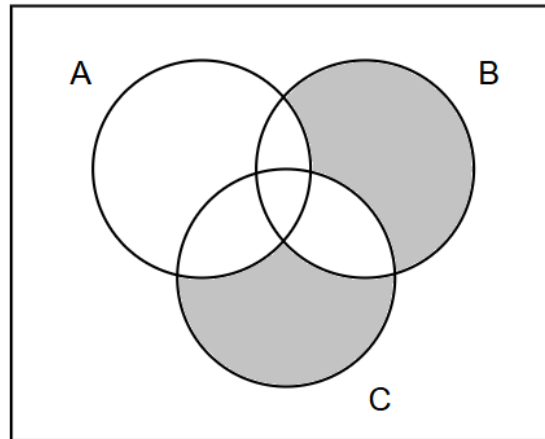
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3. “either an ace or a face card, but not both, and also not a heart”:

$$[(C \cup B) - (C \cap B)] \cap A^c.$$

$\Omega$



*Grading idea (10 pts):* correct forms (2+2+2), properly shaded Venn diagrams, (1+1+1), clear set-logic justification (1).

### Q3, 10 points

Model: draw one ball uniformly at random from 3 red, 2 blue, 1 green.

$$S = \{R_1, R_2, R_3, B_1, B_2, G\}.$$

Assume equiprobable outcomes:  $P(\omega) = \frac{1}{6} \quad \forall \omega \in S$ .

For color events,

$$\begin{aligned} P(\text{red}) &= P(\{R_1, R_2, R_3\}) = \frac{3}{6} = \frac{1}{2}, \\ P(\text{blue}) &= \frac{2}{6} = \frac{1}{3}, \\ P(\text{green}) &= \frac{1}{6}, \\ P(\text{non-red}) &= P(\text{blue} \cup \text{green}) = \frac{2}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}. \end{aligned}$$

(Check:  $P(\text{red}) + P(\text{blue}) + P(\text{green}) = 1$ .)

*Grading idea (10 pts):*  $S$  (3), probabilities on outcomes (4), non-red (3).

#### Q4, 10 points

Let  $A$  = “made in Plant A”,  $B$  = “made in Plant B”,  $D$  = “defective”. Data:

$$P(A) = 0.6, P(B) = 0.4, P(D|A) = 0.05, P(D|B) = 0.10.$$

Law of total probability:

$$\begin{aligned} P(D) &= P(D|A)P(A) + P(D|B)P(B) \\ &= (0.05)(0.6) + (0.10)(0.4) \\ &= 0.03 + 0.04 = 0.07. \end{aligned}$$

Bayes’ rule:

$$P(B|D) = \frac{P(D|B)P(B)}{P(D)} = \frac{0.10 \cdot 0.4}{0.07} = \frac{0.04}{0.07} \approx 0.571428 \dots$$

*Interpretation:* Given a defect, it is more likely from Plant B because of the higher defect rate despite lower production share.

*Grading idea (10 pts):*  $P(D)$  setup (2), Bayes application (4), numeric conclusion (2).

#### Q5, 10 points

Let  $D$  = “has disease”,  $+$  = “tests positive”.

$$\begin{aligned} P(D) &= 0.01, & P(D^c) &= 0.99, \\ P(+|D) &= 0.95, & P(+|D^c) &= 0.02. \end{aligned}$$

Total probability for a positive test:

$$\begin{aligned} P(+) &= P(+|D)P(D) + P(+|D^c)P(D^c) \\ &= (0.95)(0.01) + (0.02)(0.99) \\ &= 0.0095 + 0.0198 = 0.0293. \end{aligned}$$

Posterior via Bayes:

$$P(D|+) = \frac{P(+|D)P(D)}{P(+)} = \frac{0.95 \cdot 0.01}{0.0293} = \frac{0.0095}{0.0293} \approx 0.3239.$$

*Odds form (optional check):* posterior odds = prior odds  $\times$  likelihood ratio =  $\frac{0.01}{0.99} \times \frac{0.95}{0.02} \approx 0.477 \Rightarrow P \approx \frac{0.477}{1+0.477} \approx 0.323$ .

*Grading idea (10 pts):* total probability (4), Bayes computation (4), clear final statement (2).

#### Q6, 10 points

We are rolling a fair 4-sided die twice. The sample space has 16 equally likely outcomes.

Define the Events:

- $A = \{\text{first roll} < 3\} = \{\text{first roll is 1 or 2}\}$ . Hence,  $P(A) = \frac{2}{4} = \frac{1}{2}$ .
- $B = \{\text{second roll} > 2\} = \{\text{second roll is 3 or 4}\}$ . Hence,  $P(B) = \frac{2}{4} = \frac{1}{2}$ .
- $C = \{\text{sum of rolls} > 6\}$ . Possible sums are 2,  $\dots$ , 8. Favorable cases are:

$$(3, 4), \quad (4, 3), \quad (4, 4).$$

Thus,  $P(C) = \frac{3}{16}$ .

(a) Independence of  $A$  and  $C$ :

We need to check whether  $P(A \cap C) = P(A)P(C)$ .

If  $A$  occurs, the first roll is 1 or 2. Then the maximum possible sum is:

$$1 + 4 = 5, \quad 2 + 4 = 6,$$

which is not greater than 6. Hence,  $A \cap C = \emptyset$ .

$$P(A \cap C) = 0.$$

Meanwhile,

$$P(A)P(C) = \frac{1}{2} \cdot \frac{3}{16} = \frac{3}{32}.$$

Since  $0 \neq \frac{3}{32}$ ,  $A$  and  $C$  are **not independent**.

False

(b) Mutual Independence of  $A, B, C$

For three events to be mutually independent, all pairs must be independent.

From part (a),  $A$  and  $C$  are not independent. Hence,  $A, B, C$  are **not mutually independent**.

False

(c) Conditional Independence of  $A$  and  $B$  given  $C$

By definition,  $A$  and  $B$  are conditionally independent given  $C$  if:

$$P(A \cap B \mid C) = P(A \mid C)P(B \mid C).$$

Step 1: Compute  $P(A \mid C)$ . From earlier,  $A \cap C = \emptyset$ .

$$P(A \mid C) = \frac{P(A \cap C)}{P(C)} = \frac{0}{3/16} = 0.$$

Compute  $P(B \mid C)$ . We need outcomes where  $B$  and  $C$  both occur.

Favorable outcomes for  $C$ :  $(3, 4), (4, 3), (4, 4)$ . Among these, the second roll is  $> 2$  in all three cases.

So  $P(B \cap C) = \frac{3}{16}$ , hence:

$$P(B \mid C) = \frac{3/16}{3/16} = 1.$$

Compute  $P(A \cap B \mid C)$ . Since  $A \cap C = \emptyset$ , also  $A \cap B \cap C = \emptyset$ .

$$P(A \cap B \mid C) = \frac{0}{3/16} = 0.$$

Verify conditional independence.

$$P(A \cap B \mid C) = 0, \quad P(A \mid C)P(B \mid C) = 0 \cdot 1 = 0.$$

They are equal. Hence,  $A$  and  $B$  are **conditionally independent given  $C$** .

False (since the statement claims they are not conditionally independent)

Final Answers:

(a) False

(b) False

(c) False

*Grading idea (10 pts):* (a) (3), (b) (2), (c) (5).

### Q7 (Medium, 10 points)

We consider two rolls of a fair 6-sided die. The sample space has 36 equally likely outcomes.

Define the Events

- $A = \{\text{first roll is in } \{1, 2, 4\}\}$ . There are 3 favorable values for the first roll out of 6, hence:

$$P(A) = \frac{3}{6} = \frac{1}{2}.$$

- $B = \{\text{both rolls are the same}\}$ . Possible outcomes:  $(1, 1), (2, 2), \dots, (6, 6)$ , so there are 6 outcomes out of 36. Thus,

$$P(B) = \frac{6}{36} = \frac{1}{6}.$$

Compute  $P(A \cap B)$

For both  $A$  and  $B$  to occur, the first roll must be in  $\{1, 2, 4\}$  and both rolls must be equal.

Favorable outcomes are:

$$(1, 1), \quad (2, 2), \quad (4, 4).$$

So there are 3 outcomes.

$$P(A \cap B) = \frac{3}{36} = \frac{1}{12}.$$

Check Independence

Two events are independent if:

$$P(A \cap B) = P(A)P(B).$$

We have:

$$P(A)P(B) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}.$$

Since

$$P(A \cap B) = P(A)P(B),$$

the events  $A$  and  $B$  are **independent**.

Yes, the events are independent.

*Grading idea (10 pts):* Define events (3), compute intersection (3), checking independence (3), final statement (1).

### Q8, 10 points

Form a 4-digit string using digits  $\{1, 2, 3, 4, 5\}$  with repetition allowed. By the multiplication principle:

$$\# = 5 \times 5 \times 5 \times 5 = 5^4 = 625.$$

*Grading idea (10 pts):* setup (4), rule application (4), final count (2).

### Q9, 10 points

A password has 6 characters from the 26 uppercase English letters (A–Z). Repetition is allowed.

Each position has 26 choices, so by the multiplication principle:

$$N_{\text{total}} = 26^6.$$

Let the set of vowels be  $V = \{A, E, I, O, U\}$  (5 letters) and consonants be the other 21 letters. Count using the complement (no vowels):

$$\#\{\text{no vowels}\} = 21^6.$$

Thus

$$\#\{\text{at least one vowel}\} = 26^6 - 21^6.$$

Under the uniform model,

$$\Pr(\text{at least one vowel}) = \frac{26^6 - 21^6}{26^6} = 1 - \left(\frac{21}{26}\right)^6.$$

**Optional numeric approximation.**

$$1 - \left(\frac{21}{26}\right)^6 \approx 1 - 0.278 \approx 0.702.$$

*Grading idea (10 pts):* total count  $26^6$  (2), complement method (4), no-vowel count  $21^6$  (2), final probability (2).

### Q10, 10 points

Choose 5 people from 8 professors ( $P$ ) and 6 students ( $S$ ).

1. At least 2 students.

$$\begin{aligned}\text{Total committees} &= \binom{14}{5} = 2002. \\ \text{Count with } \geq 2 \text{ students} &= \binom{6}{2}\binom{8}{3} + \binom{6}{3}\binom{8}{2} + \binom{6}{4}\binom{8}{1} + \binom{6}{5}\binom{8}{0} \\ &= \binom{6}{2}\binom{8}{3} + \binom{6}{3}\binom{8}{2} + \binom{6}{4}\binom{8}{1} + \binom{6}{5}\binom{8}{0} \\ &= 840 + 560 + 120 + 6 = 1526.\end{aligned}$$

*Check (complement method):* (0 or 1 student)  $= \binom{6}{0}\binom{8}{5} + \binom{6}{1}\binom{8}{4} = 56 + 420 = 476$ ; then  $2002 - 476 = 1526$  (consistent).

2. Probability of exactly 3 professors and 2 students:

$$\frac{\binom{8}{3}\binom{6}{2}}{\binom{14}{5}} = \frac{840}{2002} \approx 0.41958.$$

*Grading idea (10 pts):* structured cases or complement (5), arithmetic/values (2), probability in part (2) with simplification/decimal (3).

### Q11, 10 points

If we connect any 3 dots, we'll get a unique triangle.

Since the order in which we select the dots does not matter, we can use combinations.

We can select 3 dots from 15 dots in  ${}^{15}C_3$  ways:

$${}^{15}C_3 = \frac{15!}{12! \times 3!} = \frac{15 \times 14 \times 13}{3 \times 2 \times 1} = 455$$

Some of these 455 triangles will be equilateral triangles, so we must subtract them from 455. Equilateral triangles can only be formed using three equally spaced points across the circle's circumference. Suppose the points are numbers from 1 to 15. From point 1 to point 6 is one-third of the circle — again, from point 6 to point 11, and from point 11 back to point 1. That's one equilateral triangle. We could make an equilateral triangle using points:

- {1, 6, 11}
- {2, 7, 12}
- {3, 8, 13}

- $\{4, 9, 14\}$
- $\{5, 10, 15\}$

After  $\{5, 10, 15\}$ , the dots repeat.

Compute no restrictions - undesirable = desirable:

$$\text{Final Answer} = 455 - 5 = 450$$

*Grading idea (10 pts):* reasoning for no-restrictions case (2), no-restrictions case calculation (2), reasoning for equilateral triangles (3) equilateral triangle point triplets (2), final answer (1).

## Q12, 10 points

Define the events:

- $B = \{\text{the taxi involved was blue}\}$
- $G = \{\text{the taxi involved was green}\}$
- $W = \{\text{the witness reports that the taxi was blue}\}$

Prior probabilities

Since there are 100 taxis:

$$P(B) = \frac{1}{100} = 0.01, \quad P(G) = \frac{99}{100} = 0.99.$$

Conditional probabilities of witness report

$$P(W | B) = 0.99, \quad P(W | G) = 0.02.$$

Total probability of witness reporting blue

Using the law of total probability:

$$P(W) = P(W | B)P(B) + P(W | G)P(G).$$

Substitute values:

$$P(W) = (0.99)(0.01) + (0.02)(0.99).$$

$$P(W) = 0.0099 + 0.0198 = 0.0297.$$

Bayes' Rule

We want:

$$P(B | W) = \frac{P(W | B)P(B)}{P(W)}.$$

Substitute values:

$$P(B | W) = \frac{0.99 \cdot 0.01}{0.0297}.$$

$$P(B | W) = \frac{0.0099}{0.0297} \approx 0.333.$$

Final Answer:

$$\boxed{P(B | W) \approx 0.333}$$

So, despite the witness's high accuracy rate, the probability that the taxi was actually blue is only about 33.3%, due to the overwhelming number of green taxis in the city.

*Grading idea (10 pts):* Define and compute prior and conditional probabilities of events (4), total probability calculation (3), Bayes' rule calculation (2), final answer (1).